

**Team Name:** NexShield

**Team Members:**

Rohan Karna (M25MAC005)

Hrishita Das(M25CSE015)

Sushantak Parashar Jha(M25CSA035)

**Institute:** IIT Jodhpur

## **Problem Statement**

Cyber fraud that targets SBI customers works on two levels at the same time: coordinated regional campaigns that affect thousands of users and individual-level attacks that take advantage of personal behavior patterns. The solutions we have now are only reactive, meaning they only work after money has been lost. There is no system today that can tell when fraud is coming or learn how each user behaves to catch strange behavior in real time, on their own device.

## **Proposed Solution**

We propose an AI fraud prevention system that works on two levels: it predicts threats on a country level and protects private behavior on a device. The two parts work on their own, but they are connected by a risk-context bridge that makes the system stronger than either part on its own.

### **Component 1: Fraud Wave Predictor**

Gets fraud complaint data from public cybercrime sites (I4C) and YONO's in-app reporting tool. Uses LSTM and Prophet time-series models to predict when and where the next big fraud campaign will happen. DBSCAN geospatial clustering shows high-risk areas in India in real time. Users in a district get a push alert before any scam message reaches them when the district's risk score goes above a learned threshold.

### **Component 2: On-Device Behavioral Anomaly Detector**

A quantized TensorFlow Lite model runs quietly on each user's phone inside YONO. It learns each person's normal behavior patterns, such as when, where, and how much they spend, without sending any personal information to a server. Sharp changes from the baseline cause instant alerts on the device and block suspicious transactions in real time. Federated Learning allows the global model to improve across 500 million users while zero personal data ever leaves any device.

## Risk-Context Bridge

An encrypted signal layer connects the two parts. When the macro layer sees that a fraud campaign is starting to build up in a certain area, it sends out a light signal that automatically makes every on-device model in that area more sensitive. The system is most aware of attacks happening right then and there, with no need for any user action or manual updates.

## System Architecture

